

Notes: Koijen, Van Nieuwerburgh & Yogo (JF, 2016)

Health and Mortality Delta: Assessing the Welfare Cost of Household Insurance Choice

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1 Big picture

- **Question.** How should households measure and manage the joint health and mortality exposures embedded in life insurance, annuities, private pensions, and long-term care insurance?
- **Risk-measure contribution.** The paper introduces two insurance risk measures: **health delta**, the payoff difference between poor health and good health, and **mortality delta**, the payoff difference between death and good health.
- **Why this is useful.** A life insurance contract, annuity, or long-term care policy is not a one-dimensional product. Each product loads differently on health risk and mortality risk. The two deltas provide a common language for comparing and combining all products.
- **Theory.** A life-cycle model with health transitions, mortality risk, health expenses, income, housing/financial wealth, bequest motives, and complete insurance markets implies optimal consumption, optimal health delta, and optimal mortality delta.
- **Empirical setting.** The model is calibrated and estimated using the Health and Retirement Study (HRS), focusing on males aged 51 and older from 1992 to 2010 with positive income and wealth and adequate basic health coverage.
- **Preference estimates.** Estimated relative risk aversion is 2.17. The average bequest motive is 5.20. The utility weight for poor health is 0.74 relative to 1.00 for good health, implying consumption and health are complements.
- **Observed insurance choice.** The model explains 68% of the cross-sectional variation in observed health delta and 83% of the cross-sectional variation in observed mortality delta.
- **New puzzle.** Cross-sectional insurance choices are largely rationalized, but within-household insurance choices are too persistent over time. Households fail to rebalance from life insurance toward annuities as they age.
- **Welfare cost.** For the median household aged 51 to 57, the lifetime welfare cost of market incompleteness and suboptimal insurance choice is 3.21% of total wealth. Most of this cost comes from mortality delta deviations, not health delta deviations.
- **Practical conclusion.** Financial advice should not evaluate life insurance, annuities, and long-term care insurance one at a time. It should build a portfolio that matches the household's desired health and mortality delta over the life cycle.

2 Data and measurement

2.1 HRS sample and population

- The data are from the Health and Retirement Study, a biennial panel of older U.S. households.
- The paper uses HRS waves from 1992 to 2010 and calibrates the life-cycle model so that each period equals two years.
- The empirical sample focuses on households whose primary respondent is male and age 51 or older at the interview.
- The paper requires positive income and positive financial plus housing wealth.
- The sample is restricted to households with adequate health insurance coverage so that the main residual out-of-pocket health risk is nursing home and long-term care spending.
- Households on Medicaid are excluded. For age 65 and older, adequate coverage means more than traditional Medicare Parts A and B; it includes Medicare Advantage, Part D, Medigap, or long-term care insurance.

Interpretation: The restriction improves comparability. The authors do not want the estimated health risk to reflect large differences in baseline health insurance coverage.

2.2 Health states

The model has three health states:

1. Death.
2. Poor health.
3. Good health.

Poor health is defined using both mortality risk and health spending risk. A respondent is in poor health if the predicted two-year mortality rate is above the median conditional on age and birth cohort, and out-of-pocket health expenses are above the median conditional on age, birth cohort, and long-term care insurance ownership.

Interpretation: The paper deliberately separates health risk from mortality risk. These are correlated but not identical, so the household needs two risk measures rather than one.

2.3 Mortality prediction and health expenses

- A probit model predicts death within two years using doctor-diagnosed conditions, age, age interactions, and birth-cohort dummies.
- Diabetes, cancer, lung disease, heart problems, and stroke raise predicted mortality risk.
- Health expenses include out-of-pocket medical spending, including nursing home and end-of-life expenses.
- The model abstracts from endogenous medical spending and treats out-of-pocket expenses as state-dependent shocks.

Interpretation: Health states are empirical objects constructed from HRS data. They are not just self-reported health categories; they combine medical information, mortality prediction, and realized health spending.

2.4 Insurance products measured in the HRS

The main products are:

- term life insurance;
- whole life insurance;
- annuities including private pensions;
- annuities excluding private pensions;
- long-term care insurance.

The empirical delta of a household is computed from ownership and coverage of these policies.

Interpretation: Private pensions are treated as annuity-like claims because they produce survival-contingent income. This is crucial: much annuitization in the data happens by default through pension income rather than through active annuity purchases.

2.5 Wealth and income

- Income includes household income but excludes annuities and private pensions when computing disposable income in the model.
- Financial and housing wealth equals total household assets minus liabilities.
- The model treats bonds and housing as state-independent assets with zero health and mortality delta.

Interpretation: Bonds and housing can move wealth across time, but they cannot target wealth to poor health or death states. Insurance products are valuable because their payoffs are state-contingent.

2.6 Measurement of health and mortality delta

For an insurance product $i \in \{L, A, H\}$ of maturity n in period t , health delta is

$$\Delta_{i,t}(n) = P_{i,t+1}(n-1 | 2) + D_{i,t+1}(n-1 | 2) - P_{i,t+1}(n-1 | 3) - D_{i,t+1}(n-1 | 3), \quad (1)$$

where state 2 is poor health and state 3 is good health. Mortality delta is

$$\delta_{i,t}(n) = D_{i,t+1}(n-1 | 1) - P_{i,t+1}(n-1 | 3) - D_{i,t+1}(n-1 | 3). \quad (2)$$

Interpretation: Health delta asks: how much more wealth does this product deliver in poor health than in good health? Mortality delta asks: how much more wealth does it deliver at death than in good health?

3 Model structure and equations

3.1 Health transition probabilities

Let $h_t \in \{1, 2, 3\}$ denote death, poor health, and good health. Health follows a Markov chain:

$$\pi_t(i, j) = \Pr(h_{t+1} = j \mid h_t = i). \quad (3)$$

The n -period transition probability is

$$\pi_t^n(i, j) = e_i' \left(\prod_{s=0}^{n-1} \pi_{t+s} \right) e_j. \quad (4)$$

The survival probability is

$$q_t(n \mid i) = 1 - \pi_t^n(i, 1). \quad (5)$$

Interpretation: The transition matrix lets the model price insurance contracts and compute expected lifetime utility conditional on age and health.

3.2 Insurance products

Term life insurance of maturity n pays one unit at death:

$$D_{L,t+s}(n - s \mid h_{t+s}) = \mathbf{1}_{t+s}(1). \quad (6)$$

Its price is the discounted sum of mortality probabilities:

$$P_{L,t}(n \mid h_t) = \sum_{s=1}^n \frac{p_t(s \mid h_t)}{R_L^s}. \quad (7)$$

A deferred annuity starts paying after maturity n and pays while alive:

$$D_{A,t+s}(n - s \mid h_{t+s}) = \begin{cases} 0, & s < n, \\ 1 - \mathbf{1}_{t+s}(1), & s \geq n. \end{cases} \quad (8)$$

Its price is

$$P_{A,t}(n \mid h_t) = \sum_{s=n}^{T-t} \frac{q_t(s \mid h_t)}{R_A^s}. \quad (9)$$

Supplemental health insurance covers the incremental cost of poor health:

$$D_{H,t+s}(n - s \mid h_{t+s}) = \mathbf{1}_{t+s}(2) (M_{t+s}(2) - M_{t+s}(3)). \quad (10)$$

Interpretation: Life insurance delivers wealth to death; annuities deliver wealth to survival; long-term care insurance delivers wealth to poor health. But each product also affects the other delta because payoffs and continuation values depend on health.

3.3 Budget constraint

Let A_t be financial plus housing wealth, Y_t income, M_t out-of-pocket health expenses, C_t consumption, F_t bond holdings, and $B_{i,t}(n)$ the quantity of insurance product i with maturity n . Savings satisfy

$$A_t + Y_t - M_t - C_t = \frac{F_t + G_t}{R} + \sum_{i \in \{L, A, H\}} \sum_{n=1}^{T-t} P_{i,t}(n) B_{i,t}(n). \quad (11)$$

Wealth next period is

$$A_{t+1} = F_t + G_t + \sum_{i \in \{L, A, H\}} \sum_{n=1}^{T-t} (P_{i,t+1}(n-1) + D_{i,t+1}(n-1)) B_{i,t}(n). \quad (12)$$

Interpretation: The budget constraint converts policy holdings into state-contingent future wealth. Health and mortality delta summarize exactly how wealth changes across future health states.

3.4 Preferences

The household maximizes expected utility over consumption while alive and bequests at death:

$$U_t(h_t) = \left\{ \omega(h_t)^\gamma C_t^{1-\gamma} + \beta \pi_t(h_t, 1) \omega(1)^\gamma A_{t+1}(1)^{1-\gamma} + \beta \sum_{j=2}^3 \pi_t(h_t, j) U_{t+1}(j)^{1-\gamma} \right\}^{1/(1-\gamma)}. \quad (13)$$

The parameter γ is relative risk aversion. The utility weight $\omega(1)$ captures the bequest motive. If $\omega(2) < \omega(3)$, consumption and health are complements: consumption is less valuable in poor health than in good health.

Interpretation: The model can rationalize low demand for long-term care insurance if poor health reduces the marginal utility of consumption.

3.5 Optimal health and mortality delta

Health delta for the household is

$$\Delta_t = A_{t+1}(2) - A_{t+1}(3). \quad (14)$$

Mortality delta is

$$\delta_t = A_{t+1}(1) - A_{t+1}(3). \quad (15)$$

Proposition 1 gives optimal consumption and optimal deltas. Optimal health delta is shaped by three forces:

- relative marginal utility in poor versus good health;
- differences in average propensities to consume across health states;
- differences in lifetime disposable income across health states.

Optimal mortality delta is shaped by:

- the bequest motive;
- the propensity to consume in good health;
- lifetime disposable income in good health.

Interpretation: The model does not ask whether households should buy one product. It asks what target payoff profile the household wants across poor health, good health, and death.

3.6 Portfolio replication

Proposition 2 says a portfolio of policies is optimal if it matches the optimal deltas:

$$\Delta_t^* = \sum_{i \in \{L, A, H\}} \sum_{n=1}^{T-t} \Delta_{i,t}(n) B_{i,t}(n), \quad (16)$$

$$\delta_t^* = \sum_{i \in \{L, A, H\}} \sum_{n=1}^{T-t} \delta_{i,t}(n) B_{i,t}(n). \quad (17)$$

Interpretation: Health and mortality delta are sufficient statistics for designing the insurance portfolio. Many portfolios can be optimal if they deliver the same two deltas and same budget cost.

3.7 Welfare cost

Let V_t^* be the value function under optimal deltas and V_t the value under observed or alternative deltas. The welfare cost is

$$L_t(n) = \frac{V_t}{V_t^*} - 1, \tag{18}$$

which the paper approximates with a second-order expansion in deviations from optimal health and mortality delta.

Interpretation: Deviations matter quadratically around the optimum. The welfare loss is an integrated cost of market incompleteness, product frictions, and suboptimal household choice.

4 Identification, estimation, and empirical design

4.1 What is identified

The paper estimates preference parameters and household-specific bequest motives using insurance choices. The key idea is that ownership of life insurance, annuities, pensions, and long-term care insurance reveals the household's desired wealth allocation across future health and death states.

Interpretation: Insurance choice is more informative than realized saving because insurance payoffs reveal which state of the world the household wants to transfer wealth into.

4.2 First-step preference estimation

- The model uses observed health and mortality delta implied by household insurance holdings.
- It chooses preference parameters so that model-implied optimal deltas are close to observed deltas.
- The authors estimate relative risk aversion, the average bequest motive, and the utility weight for poor health by continuous-updating GMM.
- The subjective discount factor is calibrated to 0.96 annually, and the good-health utility weight is normalized to 1.

Interpretation: The preference estimates are identified by the shape of insurance portfolios, not by consumption data alone.

4.3 Second-step household bequest motives

After estimating common preference parameters, the paper allows the bequest motive to vary across households. Estimated bequest motives are then regressed on demographics, wealth, education, marital status, children, race/ethnicity, health, age, and birth cohorts.

Interpretation: Household heterogeneity in mortality delta is mostly interpreted as heterogeneity in bequest motives. This is why the model can explain a large share of cross-sectional variation in mortality delta.

4.4 Observed versus optimal demand

The paper compares observed household deltas with model-implied optimal deltas.

- Cross-sectional fit is strong.
- Within-household life-cycle dynamics are weak: observed deltas are more persistent than the model predicts.
- The main mismatch is not simply low annuity ownership; it is insufficient rebalancing of life insurance and annuities over time.

Interpretation: The paper discovers a dynamic portfolio puzzle. Even if average annuitization is not obviously wrong once bequests are allowed, the time path of annuitization is too sticky.

4.5 Welfare interpretation

The welfare cost is the joint cost of:

- incomplete markets;
- actuarial markups;

- borrowing constraints;
- private information;
- behavioral mistakes or suboptimal choice.

Interpretation: The paper does not separately identify these frictions. It measures how costly the observed insurance portfolio is relative to the complete-markets benchmark.

4.6 Limits of the design

- The paper is structural, not a randomized experiment.
- It assumes the HRS insurance and wealth measures can be mapped into the model’s policy positions.
- It abstracts from residual private information beyond verifiable health.
- It treats basic health insurance coverage as sufficiently uniform in the selected sample.
- It estimates the welfare cost of deviations but cannot fully separate market incompleteness from household mistakes.

Interpretation: The paper’s strength is not causal treatment assignment. Its strength is a disciplined mapping from rich household data into a state-contingent portfolio model.

5 Paper roadmap: section-by-section detailed notes

5.1 Introduction

- The paper motivates health and mortality delta by analogy to beta and duration: households need compact risk measures to manage insurance portfolios.
- It argues that product-by-product analysis is misleading because products are substitutes or complements in hidden ways.
- It previews the main empirical result: a median household aged 51 to 57 bears a lifetime welfare cost of about 3.2% of total wealth.

5.2 Section I: Life-cycle model with health and mortality risk

- Sets up health transitions, out-of-pocket expenses, income, wealth, housing, and insurance products.
- Defines life insurance, annuities, and supplemental health insurance payoffs and prices.
- Defines health delta and mortality delta for each product.
- Shows that long-term life insurance and supplemental health insurance have positive health delta, while deferred annuities have negative health delta.
- Shows that life insurance has positive mortality delta, while annuities and long-term care insurance have negative mortality delta.

5.3 Section II: Optimal demand for insurance

- Characterizes the complete-markets solution using optimal consumption and optimal deltas.
- Proposition 1 derives optimal health delta and mortality delta.
- Proposition 2 shows that any portfolio of products that replicates the optimal deltas is optimal.
- Proposition 3 gives the second-order approximation used to measure welfare cost.

5.4 Section III: Calibrating the life-cycle model

- Describes the HRS sample and health-state definition.
- Uses mortality prediction and out-of-pocket health expenses to define poor health.
- Estimates health transition probabilities.
- Estimates out-of-pocket health expenses and income profiles by age and health.
- Computes household deltas from observed insurance coverage.

5.5 Section IV: Assessing household insurance choice

- Estimates preference parameters and household-specific bequest motives.
- Reports that bequest motives are stronger for married households, households with children, more educated households, and wealthier households.
- Compares observed deltas with optimal deltas.
- Shows strong cross-sectional fit but weak within-household time variation.
- Computes per-period and lifetime welfare costs.

5.6 Section V: Optimal portfolio of existing products

- Demonstrates how short-term life insurance, deferred annuities, long-term care insurance, and bonds can replicate the model-implied optimal deltas.
- The illustrative household starts with average wealth and income and is a male born 1936-1940 in good health at age 51.
- The optimal portfolio rebalances from large life-insurance positions at age 51 toward annuities later in life.

5.7 Section VI and appendices

- The conclusion emphasizes financial advice: households should manage the health and mortality delta of their entire insurance portfolio.
- Appendix A proves the propositions.
- Appendix B describes the construction of health expenses, income, life insurance, annuities, and wealth.
- Appendix C checks robustness under actuarially unfair insurance, longer term-life maturity, and an alternative poor-health definition.
- Appendix D models future transitions in insurance ownership for lifetime welfare-cost calculations.

6 Figures and tables

6.1 Figure 1 and Figure 2: What the risk measures mean and how health evolves

Read:

- Figure 1 defines the economics of health and mortality delta. Health delta compares poor-health payoffs to good-health payoffs. Mortality delta compares death-state payoffs to good-health payoffs.
- Life insurance is high mortality-delta insurance. Deferred annuities are negative mortality-delta products because they pay only when alive.
- Long-term care insurance has positive health delta because it pays in poor health, but it can also have negative mortality delta because its value is lost at death.
- Figure 2 shows health transition probabilities by age, current health, and birth cohort. Death risk rises with age and poor health; older cohorts have worse transitions.

Economic message: The deltas convert a complicated menu of insurance products into two state-contingent payoff slopes. Figure 2 supplies the transition probabilities needed to price those slopes.

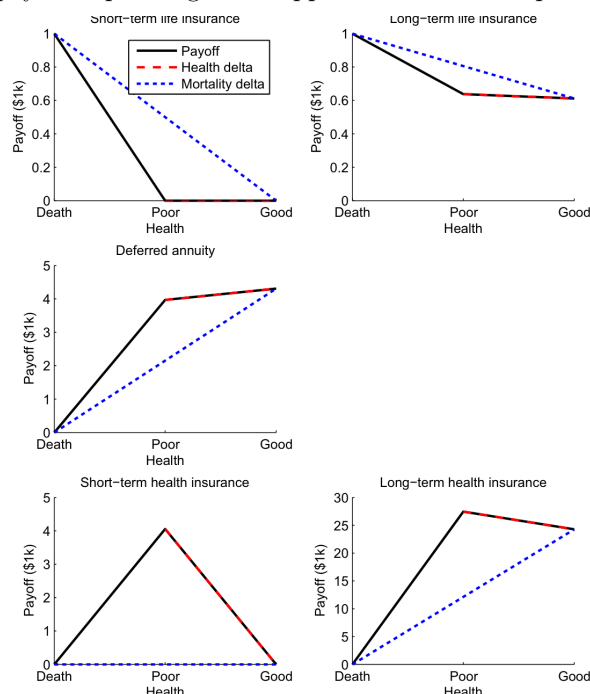


Figure 1. Health and mortality delta for insurance products. The solid line represents the payoff of each policy for the three possible health states. The dashed and dotted lines represent the health and mortality deltas, respectively.

Figure 1: product payoffs and deltas

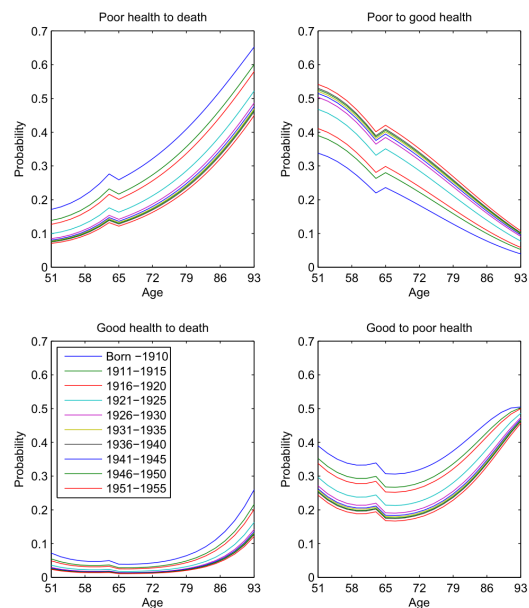


Figure 2. Estimated health transition probabilities. An ordered probit model is used to predict the health state at two years from the present interview. The explanatory variables are dummies for present health and 65 or older, a quadratic polynomial in age, log income, the in-

Figure 2: estimated health transition probabilities

6.2 Tables I and II: Health-state construction and descriptive facts

Read:

- Table I estimates a probit model for death within two years. Cancer raises mortality risk by 10.43 percentage points, lung disease by 6.07, diabetes by 4.49, stroke by 3.62, and heart problems by 1.98.
- The model correctly predicts 94% of both outcomes and 66% of deaths, which is important because the poor-health definition uses predicted mortality.
- Table II validates the poor-health definition. Poor-health respondents have more diagnosed conditions, more health care use, and higher nursing-home use.

- For ages 79 and older, nursing-home stays are 14% for poor health versus 5% for good health.
- Insurance ownership varies by age. Annuity and pension ownership rises with age, while term-life coverage falls.

Economic message: The paper’s health states have real content: poor health predicts both mortality and medical spending risk. The state definition is therefore suitable for insurance portfolio analysis.

Table I
Predicting Future Mortality with Observed Health Problems

A probit model is used to predict death within two years of the present interview. This table reports the marginal effects on the mortality rate (in percentage points) with heteroskedasticity-robust t -statistics in parentheses. The sample consists of males aged 51 and older in the Health and Retirement Study from 1992 to 2010. The omitted cohort consists of males born prior to 1911.

Explanatory Variable	Marginal Effect	t -Statistic
Doctor-diagnosed health problems:		
High blood pressure	0.70	(1.64)
Diabetes	4.49	(5.32)
Cancer	10.43	(7.10)
Lung disease	6.07	(4.57)
Heart problems	1.98	(3.39)
Stroke	3.62	(2.84)
(Age = 51)/10	2.28	(11.16)
× High blood pressure	−0.03	(−0.18)
× Diabetes	−0.61	(−2.85)
× Cancer	−1.39	(−6.29)
× Lung disease	0.03	(0.11)
× Heart problems	0.08	(0.44)
× Stroke	−0.02	(−0.08)
Birth cohort:		
1911–1915	−1.24	(−3.77)
1916–1920	−1.83	(−6.73)
1921–1925	−2.56	(−10.94)
1926–1930	−3.02	(−12.63)
1931–1935	−3.34	(−10.54)
1936–1940	−3.62	(−9.38)
1941–1945	−3.11	(−10.29)
1946–1950	−3.20	(−13.49)
1951–1955	−2.84	(−9.99)
Correctly predicted (%):		
Both outcomes	94	
Death only	66	
Alive only	94	
Observations	38,913	

Mortality rates and health expenses are not perfectly correlated, and they both contain important information about true unobserved health. We therefore define three health states based on both factors.

1. Death.
2. Poor health: The predicted mortality rate is higher than the median conditional on age and birth cohort. In addition, out-of-pocket health expenses are higher than the median conditional on age, birth cohort, and the ownership of long-term care insurance.
3. Good health: Alive and not in poor health.

Table I: predicting mortality

Table II
Health Problems, Health Care Utilization, and Insurance Ownership

Term life insurance refers to individual and group policies that have only a death benefit. Whole life insurance refers to policies that build cash value, from which the policyholder can borrow or receive cash upon surrender. The sample consists of males aged 51 and older in the Health and Retirement Study from 1992 to 2010.

Age	51–64		65–78		79–	
Health	Poor	Good	Poor	Good	Poor	Good
Panel A: Doctor-Diagnosed Health Problems and Difficulty with Activities of Daily Living (%)						
High blood pressure	57	30	67	47	65	47
Diabetes	20	8	35	14	28	15
Cancer	8	3	28	11	33	21
Lung disease	7	3	19	6	22	7
Heart problems	23	9	54	22	74	35
Stroke	5	2	15	5	31	10
Some difficulty bathing	3	1	5	2	21	9
Some difficulty dressing	6	3	10	5	24	13
Some difficulty eating	1	0	3	1	12	4
Panel B: Health Care Utilization (%)						
Monthly doctor visits	9	3	16	6	21	11
Hospital stay	25	11	42	23	55	34
Outpatient surgery	22	16	28	23	27	26
Nursing home stay	0	0	2	1	14	5
Home health care	3	1	10	5	22	10
Special facilities and services	8	4	11	6	16	10
Prescription drugs	80	51	95	77	97	86
Panel C: Insurance Ownership Rate (%)						
Term life insurance	72	71	57	59	48	48
Whole life insurance	35	34	32	31	29	28
Annuities including private pensions	45	48	55	59	58	63
Annuities excluding private pensions	1	1	4	4	6	7
Long-term care insurance	8	9	18	20	18	18
Panel D: Insurance Coverage Conditional on Ownership (median in thousands of 2005 dollars)						
Term life insurance	78.4	81.0	22.8	23.2	10.7	10.0
Whole life insurance	42.8	42.8	23.2	23.3	16.6	15.0
Annual annuity and pension income	0.0	0.0	11.9	12.8	9.0	9.4
Financial and housing wealth	152.8	175.7	233.9	257.6	234.2	244.3
Observations	7,702	12,234	4,672	10,228	1,717	3,796

D. Health and Mortality Risk

D.1. Health Transition Probabilities

After defining the three health states, we estimate the transition probabilities between the health states using an ordered probit model. The outcome

Table II: health, utilization, and insurance ownership

6.3 Table III and Figure 3: Health dynamics and late-life spending risk

Read:

- Table III reports model-implied health dynamics, life expectancy, out-of-pocket expenses, income, and the present value of future disposable income.
- For males born 1936-1940 and in good health at age 51, the probability of death by age 65 is 0.24, and the probability of poor health at age 65 is 0.22.
- Remaining life expectancy at age 65 is 17 years in poor health and 20 years in good health.
- Out-of-pocket expenses rise sharply in old age. At age 93, poor-health expenses are \$22.1k per year versus \$3.6k in good health.
- Figure 3 shows a highly skewed lifetime health-expense distribution with a long right tail exceeding \$250k.

Economic message: Self-insuring late-life medical risk requires a large buffer stock of wealth. State-contingent insurance can target that tail more efficiently.

Table III
Health Dynamics, Out-of-Pocket Health Expenses, and Income

Panels A and B are based on the estimated model for health transition probabilities. Panels C and D are based on the estimated models for out-of-pocket health expenses and income, respectively. Panel E reports the present value of future disposable income, based on the estimated health transition probabilities and a riskless interest rate of 2%. The reported estimates are for males born between 1936 and 1940, with average income and in good health at age 51.

Health	Age						
	51	58	65	72	79	86	93
Panel A: Long-Run Health Transition Probabilities							
Death	0.00	0.09	0.24	0.34	0.50	0.65	0.87
Poor	0.00	0.25	0.22	0.18	0.17	0.15	0.08
Good	1.00	0.65	0.54	0.48	0.34	0.20	0.05
Panel B: Remaining Life Expectancy (years)							
Poor	26	22	17	13	9	6	4
Good	27	24	20	17	12	9	6
Mean	27	24	19	16	11	8	5
Panel C: Out-of-Pocket Health Expenses (thousands of 2005 dollars per year)							
Poor	2.3	2.9	3.5	4.0	6.1	9.6	22.1
Good	0.4	0.7	1.0	1.2	1.7	2.2	3.6
Mean	0.4	1.3	1.7	2.0	3.1	5.4	15.0
Panel D: Income (thousands of 2005 dollars per year)							
Mean	56.1	48.4	30.6	25.2	20.7	18.8	17.7
Panel E: Present Value of Future Disposable Income (thousands of 2005 dollars)							
Poor	568.5	390.7	230.3	146.7	63.5	19.0	-23.8
Good	606.1	437.8	278.5	191.3	96.7	37.8	-30.1
Mean	606.1	424.6	264.4	179.3	85.7	29.7	-26.2

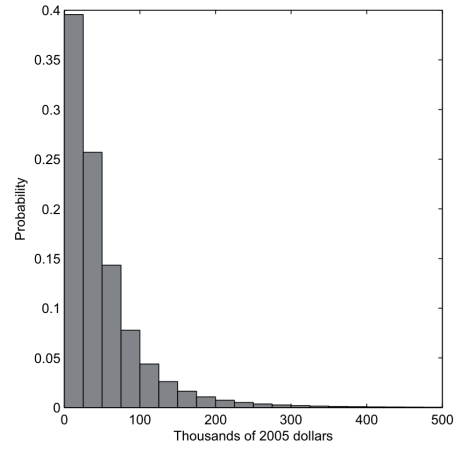


Figure 3. Lifetime out-of-pocket health expenses. The estimated models for health transition probabilities and out-of-pocket health expenses are used to simulate paths of out-of-pocket health

Table III: health dynamics, expenses, and income

Figure 3: lifetime out-of-pocket expenses

6.4 Figure 4 and Table IV: Observed deltas and estimated preferences

Read:

- Figure 4 shows observed health delta and mortality delta over the life cycle.
- Average health delta is negative for much of the life cycle because many households hold annuities or pensions but few hold long-term care insurance.
- Average mortality delta declines with age as life insurance becomes less prominent and annuity/pension claims become more important.
- Table IV estimates relative risk aversion at 2.17, the average bequest motive at 5.20, and the poor-health utility weight at 0.74.
- Since the good-health utility weight is normalized to 1.00, $\omega(2) = 0.74 < \omega(3) = 1.00$ implies consumption and health are complements.

Economic message: Insurance portfolios reveal preferences. Low long-term care demand, together with annuity/pension ownership, supports a low marginal utility of consumption in poor health.

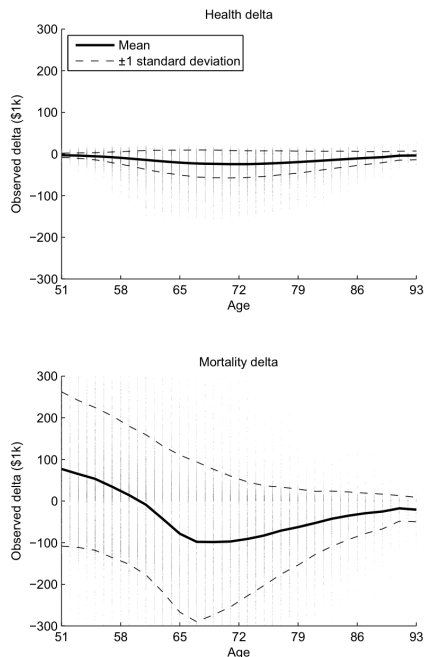


Figure 4. Observed health and mortality delta over the life cycle. Each dot in the top

Figure 4: observed deltas over the life cycle

Table IV
Estimated Household Preferences

The subjective discount factor is calibrated to 0.96 annually, and the utility weight for good health is normalized to one. The remaining preference parameters are estimated by continuous-updating generalized method of moments with heteroskedasticity-robust standard errors in parentheses. The sample consists of males aged 51 and older in the Health and Retirement Study from 1992 to 2010.

Parameter	Symbol	Value
Subjective discount factor	β	0.96
Relative risk aversion	γ	2.17 (0.01)
Average bequest motive	$\omega(1)$	5.20 (0.03)
Utility weight for poor health	$\omega(2)$	0.74 (0.01)
Utility weight for good health	$\omega(3)$	1.00
Observations		28,828

Table IV: preference estimates

Table IV: preference estimates

6.5 Figure 5 and Table V: Bequest motives

Read:

- Figure 5 reports the distribution of estimated household bequest motives, showing large heterogeneity.
- Table V explains the logarithm of estimated bequest motives using household characteristics.
- Married households have a coefficient of 0.58; households with living children have 0.29.
- Education is strongly related to bequest motives: high school graduate has 0.62, college graduate has 1.44.
- Log financial and housing wealth has coefficient 0.53, so bequest motives rise strongly with wealth.
- The regression has $R^2 = 0.66$ and 28,464 observations.

Economic message: The model's large cross-sectional fit comes partly from allowing heterogeneity in bequest motives. Family structure, education, and wealth explain much of that heterogeneity.

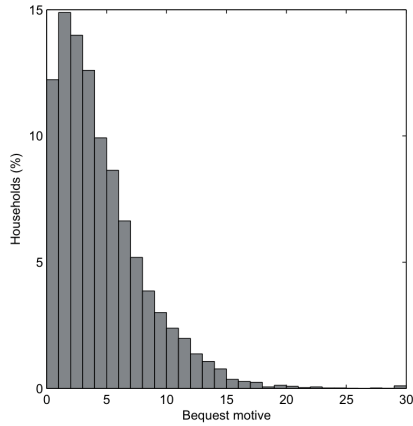


Figure 5. Estimated bequest motives. The bequest motive is estimated for each household to minimize the average per-period welfare cost. The sample consists of males aged 51 and older in

Figure 5: estimated bequest motives

Table V
Explaining the Bequest Motive

A linear regression model is estimated to explain the logarithm of estimated bequest motives. This table reports the coefficients with heteroskedasticity-robust t -statistics in parentheses. The sample consists of males aged 51 and older in the Health and Retirement Study from 1992 to 2010. The omitted categories for the dummies are no high school degree, white, good self-reported health, and born prior to 1911.

Explanatory Variable	Coefficient	t -Statistic
65 or older	0.14	(2.19)
Poor health	-0.01	(-0.30)
Married	0.58	(11.95)
Has living children	0.29	(5.03)
High school graduate	0.62	(10.57)
× Married	0.05	(1.64)
× Has living children	-0.16	(-3.77)
College graduate	1.44	(23.42)
× Married	-0.14	(-3.83)
× Has living children	-0.16	(-3.65)
Black	-0.48	(-10.71)
Hispanic and other	-0.30	(-5.68)
Self-reported health status:		
Poor	0.33	(10.92)
Fair	0.20	(7.30)
Very good	-0.16	(-3.96)
Excellent	-0.48	(-6.67)
(Age - 51)/10	0.64	(6.92)
× 65 or older	-0.22	(-2.63)
× Poor health	0.02	(0.62)
× Married	-0.10	(-2.42)
× Has living children	-0.06	(-1.31)
× High school graduate	-0.14	(-3.65)
× College graduate	-0.34	(-8.07)
× Black	0.01	(0.25)
× Hispanic and other	-0.07	(-1.04)
× Poor	-0.13	(-3.56)
× Fair	0.01	(0.20)
× Very good	0.02	(0.35)
× Excellent	0.12	(1.61)
(Age - 51) ² /100	-0.06	(-1.26)
× 65 or older	0.08	(1.71)
× Poor health	-0.01	(-1.32)
× Married	0.01	(0.72)
× Has living children	-0.01	(-0.69)
× High school graduate	0.03	(2.82)
× College graduate	0.05	(4.58)
× Black	-0.02	(-1.59)
× Hispanic and other	0.01	(0.59)
× Poor	0.02	(1.95)
× Fair	-0.02	(-2.19)
× Very good	0.00	(0.24)
× Excellent	-0.01	(-0.45)

(Continued)

Table V: bequest regression, part 1

Table V—Continued

Explanatory Variable	Coefficient	t -Statistic
Log financial and housing wealth	0.53	(21.29)
× 65 or older	0.02	(1.55)
× Poor health	-0.01	(-0.85)
× Married	0.01	(0.97)
× Has living children	-0.05	(-2.69)
× High school graduate	-0.05	(-3.85)
× College graduate	-0.06	(-4.28)
× Black	-0.05	(-2.50)
× Hispanic and other	0.06	(2.74)
× Poor	0.00	(0.18)
× Fair	-0.02	(-1.85)
× Very good	-0.01	(-0.33)
× Excellent	0.01	(0.58)
Birth cohort:		
1911–1915	0.17	(5.12)
1916–1920	0.25	(7.58)
1921–1925	0.49	(15.22)
1926–1930	0.47	(14.14)
1931–1935	0.54	(15.78)
1936–1940	0.49	(14.23)
1941–1945	0.52	(14.43)
1946–1950	0.61	(16.58)

6.6 Figure 6 and Table VI: Observed versus optimal insurance demand

Read:

- Figure 6 plots observed deltas against optimal deltas for each household-interview observation.
- In the health-delta panel, points above the 45-degree line indicate too much whole life or long-term care exposure; points below indicate too much annuity/pension exposure relative to the model.

- In the mortality-delta panel, points above the 45-degree line are under-annuitized; points below are over-annuitized.
- Table VI uses household fixed effects to explain deviations from optimal deltas.
- Age has a positive and significant effect on mortality-delta deviations, implying the model fails to explain within-household dynamics.
- The R^2 values are 0.20 for health-delta deviations and 0.15 for mortality-delta deviations.

Economic message: The model explains cross-sectional insurance positions well but not life-cycle rebalancing. The main puzzle is the persistence of observed deltas over time.

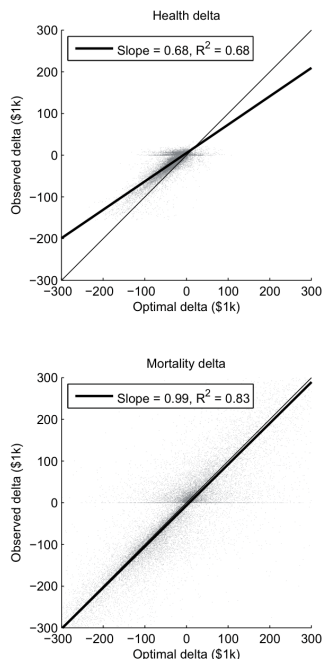


Figure 6: Observed versus optimal health and mortality delta. The optimal health and mortality deltas are predicted by the life-cycle model with household-specific bequest motives.

Table VI
Explaining the Deviations from Optimal Health and Mortality Delta

A panel regression model with household fixed effects is estimated to explain the deviations from optimal health and mortality delta. The optimal health and mortality delta are predicted by the life-cycle model with household-specific bequest motives. This table reports the coefficients with heteroskedasticity-robust t -statistics in parentheses. The sample consists of males aged 51 and older in the Health and Retirement Study from 1992 to 2010.

Explanatory Variable	Health Delta		Mortality Delta	
65 or older	-47.15	(-19.88)	-32.75	(-1.94)
Poor health	3.93	(2.76)	22.87	(2.04)
(Age - 51)/10	2.13	(1.81)	200.15	(18.45)
× 65 or older	8.98	(5.33)	-163.58	(-12.62)
× Poor health	2.25	(4.19)	7.60	(1.73)
(Age - 51) ² /100	1.89	(2.55)	-77.49	(-11.48)
× 65 or older	-4.10	(-5.22)	77.64	(11.33)
× Poor health	-0.59	(-4.16)	-2.11	(-2.07)
Log financial and housing wealth	-1.61	(-11.92)	-4.12	(-4.11)
× 65 or older	3.89	(23.42)	8.85	(6.99)
× Poor health	-0.44	(-3.54)	-2.08	(-2.15)
R^2	0.20		0.15	

Figure 6: observed versus optimal deltas

Table VI: deviations from optimal deltas

6.7 Table VII and Table VIII: Welfare costs and optimal portfolios

Read:

- Table VII reports the welfare cost of household insurance choice.
- For ages 51-57, the median lifetime welfare cost is 3.21% of total wealth.
- The cost due to health delta is 0.46%, while the cost due to mortality delta is 3.52%; the interaction term makes the total lower than their sum.
- Welfare costs decline with age because the remaining lifetime horizon shortens: total cost is 0.38% for ages 86 and older.
- Table VIII shows an optimal product portfolio for a representative male born 1936-1940.
- At age 51, optimal mortality delta is \$268.3k and optimal life-insurance units are 293.1. By age 93, life insurance is zero, deferred annuity units are 32.7, and mortality delta is -\$86.0k.

Economic message: The welfare cost is economically large and mostly driven by mortality-delta mistakes. The optimal portfolio involves rebalancing from life insurance toward annuities over the life cycle.

Table VII
Welfare Cost of Household Insurance Choice

The per-period welfare cost is based on the deviations from optimal health and mortality delta. The lifetime welfare cost is based on the predicted path of future health and mortality delta, described in Appendix D. The sample consists of males aged 51 and older in the Health and Retirement Study from 1992 to 2010.

	Age					
	51-57	58-64	65-71	72-78	79-85	86-
Panel A: Per-Period Welfare Cost (median in % of total wealth)						
Total cost	0.03 (0.00)	0.03 (0.00)	0.03 (0.00)	0.02 (0.00)	0.02 (0.00)	0.02 (0.15)
Cost due to health delta	0.01 (0.00)	0.01 (0.01)	0.02 (0.00)	0.01 (0.00)	0.01 (0.00)	0.01 (0.26)
Cost due to mortality delta	0.01 (0.01)	0.01 (0.01)	0.01 (0.00)	0.01 (0.01)	0.01 (0.01)	0.01 (0.18)
Panel B: Lifetime Welfare Cost (median in % of total wealth)						
Total cost	3.21 (0.27)	2.91 (0.19)	2.70 (0.22)	1.57 (0.22)	0.72 (0.26)	0.38 (0.49)
Cost due to health delta	0.46 (0.07)	0.38 (0.04)	0.39 (0.05)	0.23 (0.06)	0.13 (0.09)	0.15 (0.39)
Cost due to mortality delta	3.52 (0.30)	3.14 (0.21)	2.98 (0.23)	1.69 (0.24)	0.73 (0.28)	0.32 (0.51)

consumption. To put our estimate into perspective, Calvet, Campbell, and Sodin (2007) find that the welfare cost of underdiversification in stock and mutual fund portfolios is 0.51% of disposable income for the median Swedish household. Through equation (30) for $n = T - t$, we can decompose this welfare cost into the sum of three parts. The deviations from optimal health delta account for 0.46% of the welfare cost, while the deviations from optimal mortality delta account for 3.52%. The interaction between health and mortality delta explains the remainder of the welfare cost, which is -0.77% . The lifetime welfare cost is higher for younger households, for which the per-period welfare cost accumulates over a longer expected lifetime.

Table VII: welfare cost

Table VIII
Optimal Portfolio of Insurance Products

Panel A reports the optimal health and mortality delta predicted by the life-cycle model with the preference parameters in Table IV. Panel B reports a portfolio of short-term life insurance, deferred annuities, long-term care insurance (i.e., short-term supplemental health insurance), and bonds that replicates the optimal health and mortality delta. Short-term policies mature in two years, and the income from deferred annuities starts at age 65. Panel C reports the cost of the optimal portfolio in thousands of 2005 dollars, averaged across the health distribution at the given age. The reported estimates are for males born between 1936 and 1940, with average income and in good health at age 51.

	Age						
	51	58	65	72	79	86	93
Panel A: Optimal Health and Mortality Delta (thousands of 2005 dollars)							
Health delta	-1.9	-5.9	-12.9	-16.4	-17.6	-14.7	6.3
Mortality delta	268.3	132.1	27.3	-9.1	-39.4	-58.6	-86.0
Panel B: Optimal Portfolio (units)							
Life insurance	293.1	175.8	106.8	69.5	26.9	0.0	0.0
Deferred annuities	5.8	7.9	10.1	11.9	13.5	15.4	32.7
Long-term care insurance	0.0	0.0	0.0	0.0	0.0	0.1	0.6
Bonds	68.1	165.1	208.6	228.2	248.6	260.0	240.6
Panel C: Cost of the Optimal Portfolio (thousands of 2005 dollars)							
Life insurance	7.1	6.4	4.7	3.6	2.4	0.0	0.0
Deferred annuities	22.8	38.9	69.7	67.3	52.3	40.8	44.8
Long-term care insurance	0.0	0.0	0.0	0.0	0.0	0.9	13.0
Bonds	65.5	158.6	200.5	219.3	239.0	249.9	231.3
Total cost	95.4	203.9	274.9	290.2	293.6	291.6	289.1

insurance (i.e., short-term supplemental health insurance). Figure 1 reports the health and mortality delta for these insurance products at age 51. The household's preference parameters are those that we estimate in the Health and Retirement Study, reported in Table IV.

Panel A of Table VIII reports the optimal health and mortality delta, which

Table VIII: optimal insurance portfolio

6.8 Figure 7 and robustness tables: Optimal deltas over the life cycle

Read:

- Figure 7 plots optimal health and mortality delta over the life cycle, using the optimal portfolio of existing products.
- The optimal health delta is slightly negative early in retirement and becomes positive only very late in life.
- The optimal mortality delta falls steeply with age, reflecting declining desired death-state wealth and the growing role of annuities.
- Appendix Table C.I makes insurance actuarially unfair. The lifetime welfare cost for ages 51-57 rises to 3.59%, close to the benchmark 3.21%.
- Appendix Table C.II assumes longer term-life maturity. The age 51-57 lifetime cost remains 3.22%, nearly identical to the benchmark.
- Appendix Table C.III uses an alternative poor-health definition. The age 51-57 lifetime cost rises to 4.11%, suggesting the benchmark is conservative.

Economic message: The optimal path of deltas is dynamic. Robustness checks preserve the central conclusion: welfare costs are large, especially for younger older households, and mortality-delta deviations dominate.

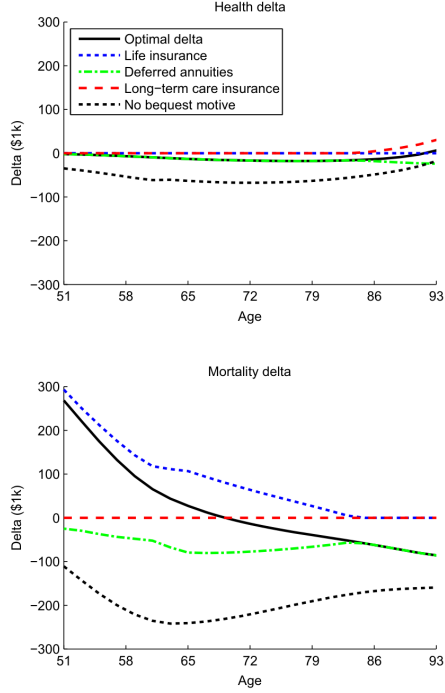


Figure 7: optimal deltas over the life cycle This figure reports the optimal health and mortality delta predicted by the life-cycle model with the preference parameters in Table IV. The sum of health (mortality) delta for short-term life insurance, deferred annuities,

Figure 7: optimal deltas over the life cycle

Table C.II
Welfare Cost of Household Insurance Choice under Term Life Insurance with Longer Maturity

This table reports the welfare cost of household insurance choice under an alternative assumption that term life insurance matures at age 65 for households younger than 65. For households older than 65, term life insurance matures in two years. The sample consists of males aged 51 and older in the Health and Retirement Study from 1992 to 2010.

	Age					
	51-57	58-64	65-71	72-78	79-85	86-
Panel A: Per-Period Welfare Cost (median in % of total wealth)						
Total cost	0.03 (0.00)	0.03 (0.00)	0.02 (0.00)	0.02 (0.00)	0.01 (0.00)	0.02 (0.15)
Cost due to health delta	0.01 (0.00)	0.02 (0.00)	0.01 (0.00)	0.01 (0.00)	0.01 (0.00)	0.01 (0.25)
Cost due to mortality delta	0.01 (0.00)	0.01 (0.00)	0.01 (0.00)	0.01 (0.00)	0.00 (0.01)	0.01 (0.18)
Panel B: Lifetime Welfare Cost (median in % of total wealth)						
Total cost	3.22 (0.27)	2.87 (0.19)	2.74 (0.22)	1.60 (0.23)	0.71 (0.27)	0.39 (0.50)
Cost due to health delta	0.43 (0.07)	0.34 (0.04)	0.34 (0.04)	0.20 (0.06)	0.13 (0.09)	0.16 (0.40)
Cost due to mortality delta	3.53 (0.30)	3.14 (0.21)	3.03 (0.24)	1.71 (0.24)	0.73 (0.28)	0.33 (0.51)

Table C.II: longer term-life maturity

Table C.I
Welfare Cost of Household Insurance Choice under Actuarially Unfair Insurance

This table reports the welfare cost of household insurance choice under an alternative assumption that insurance is more expensive than actuarially fair. The discount rates on life insurance, annuities, and long-term care insurance are calibrated to 0% annually, while the riskless interest rate is 2%. The sample consists of males aged 51 and older in the Health and Retirement Study from 1992 to 2010.

	Age					
	51-57	58-64	65-71	72-78	79-85	86-
Panel A: Per-Period Welfare Cost (median in % of total wealth)						
Total cost	0.03 (0.00)	0.03 (0.00)	0.03 (0.00)	0.02 (0.00)	0.02 (0.00)	0.02 (0.15)
Cost due to health delta	0.01 (0.00)	0.01 (0.00)	0.02 (0.00)	0.01 (0.00)	0.01 (0.00)	0.01 (0.25)
Cost due to mortality delta	0.01 (0.00)	0.01 (0.00)	0.01 (0.00)	0.01 (0.00)	0.00 (0.01)	0.01 (0.16)
Panel B: Lifetime Welfare Cost (median in % of total wealth)						
Total cost	3.59 (0.27)	3.24 (0.20)	3.03 (0.22)	1.76 (0.23)	0.79 (0.27)	0.40 (0.48)
Cost due to health delta	0.53 (0.06)	0.45 (0.04)	0.45 (0.03)	0.25 (0.04)	0.14 (0.08)	0.16 (0.38)
Cost due to mortality delta	3.99 (0.30)	3.59 (0.22)	3.40 (0.24)	1.92 (0.25)	0.80 (0.29)	0.35 (0.50)

Table C.I: actuarially unfair insurance

Table C.III
Welfare Cost of Household Insurance Choice under an Alternative Definition of Poor Health

This table reports the welfare cost of household insurance choice under an alternative assumption that poor health is a state in which out-of-pocket health expenses are higher than the 75th percentile conditional on age, birth cohort, and the ownership of long-term care insurance. The criterion for the predicted mortality rate remains at higher than the median conditional on age and birth cohort. The sample consists of males aged 51 and older in the Health and Retirement Study from 1992 to 2010.

	Age					
	51-57	58-64	65-71	72-78	79-85	86-
Panel A: Per-Period Welfare Cost (median in % of total wealth)						
Total cost	0.03 (0.00)	0.03 (0.00)	0.03 (0.00)	0.03 (0.00)	0.02 (0.01)	0.06 (0.37)
Cost due to health delta	0.01 (0.00)	0.01 (0.00)	0.02 (0.00)	0.01 (0.00)	0.01 (0.01)	0.03 (0.38)
Cost due to mortality delta	0.01 (0.00)	0.01 (0.00)	0.01 (0.00)	0.01 (0.00)	0.01 (0.01)	0.01 (0.31)
Panel B: Lifetime Welfare Cost (median in % of total wealth)						
Total cost	4.11 (0.30)	3.84 (0.23)	3.68 (0.26)	2.29 (0.29)	1.11 (0.34)	0.83 (0.67)
Cost due to health delta	0.44 (0.05)	0.41 (0.04)	0.47 (0.04)	0.33 (0.07)	0.25 (0.14)	0.39 (0.59)
Cost due to mortality delta	4.03 (0.31)	3.67 (0.23)	3.45 (0.26)	2.03 (0.27)	0.84 (0.31)	0.31 (0.56)

Appendix D: Transition Probabilities for Insurance Ownership

Table C.III: alternative poor-health definition

7 Short summary

- The paper develops health delta and mortality delta as portfolio risk measures for household insurance products.
- Health delta measures the payoff difference between poor health and good health. Mortality delta measures the payoff difference between death and good health.
- Life insurance, annuities, pensions, and long-term care insurance are substitutes and complements across these two deltas.
- A life-cycle model with complete markets gives optimal consumption, optimal health delta, and optimal mortality delta.
- The HRS provides the data for health transitions, mortality prediction, health expenses, income, wealth, and insurance ownership.
- Estimated relative risk aversion is 2.17, the average bequest motive is 5.20, and the poor-health utility weight is 0.74.
- The model explains most cross-sectional variation in observed deltas but fails to explain within-household persistence.
- The main dynamic puzzle is insufficient rebalancing from life insurance to annuities over the life cycle.
- For the median household aged 51-57, the lifetime welfare cost is 3.21% of total wealth.
- Most welfare cost comes from mortality-delta deviations, not health-delta deviations.
- The optimal portfolio illustrates a practical implementation: use existing products to match the target health and mortality delta.

8 Conclusion

8.1 Core conclusion

Koijen, Van Nieuwerburgh, and Yogo show that the household insurance problem can be summarized by two risk measures: health delta and mortality delta. These measures turn a complicated menu of life insurance, annuities, pensions, and long-term care insurance into a state-contingent portfolio problem.

8.2 Economic intuition

The household wants different wealth in different future states. Life insurance moves wealth to death. Annuities move wealth to survival. Long-term care insurance moves wealth to poor health. The right portfolio depends on how much the household values consumption in poor health, how strong the bequest motive is, how health transitions evolve, and how health expenses and income change with age.

8.3 Takeaway

The key lesson is that insurance choices should be evaluated jointly, not product by product. Many households choose portfolios whose implied mortality delta and health delta are too persistent over time. The resulting welfare cost is large relative to familiar household-finance mistakes. A practical advisory system should compute the household's target health and mortality deltas and then recommend a portfolio of existing insurance products that replicates them.